

# Decomposition of Concentration-Index Trees

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## 1 4. Results

The results section reports only the objects needed to answer the main technical question: whether inequality-oriented recursive partitioning identifies subgroup structure that is visible, stable, and comparable with classical decomposition output. Full generated outputs are moved to the appendix.

### 1.1 4.1 Exploratory Data Analysis



Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

### 1.2 4.2 Application of developed tree models to the DRC DHS Data

#### 1.2.1 4.2.1 Comparison of Indirect Decomposition Results

Categorical predictors enter the regression decomposition as separate one-hot terms. The predictive ranger is fitted on the same design-matrix terms, so the comparison table reports categorical levels at the same granularity instead of collapsing them back to their parent variable.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion

| Regression | Predictive-forest SHAP |
|------------|------------------------|
|------------|------------------------|

| Variable                          | CI (%) | CIg (%) | CIc (%) | CI (%) | CIg (%) | CIc (%) | L (%)  |
|-----------------------------------|--------|---------|---------|--------|---------|---------|--------|
| Residence                         | 33.993 | 70.954  | 70.954  | 32.349 | 32.349  | 32.349  | 38.158 |
| Province: Lualaba                 | 6.713  | 1.192   | 1.192   | -5.373 | -5.373  | -5.373  | -4.558 |
| Province: Sankuru                 | 6.700  | 1.570   | 1.570   | 3.670  | 3.670   | 3.670   | 1.469  |
| Partner occupation: d Agriculture | 6.406  | 7.369   | 7.369   | 7.770  | 7.770   | 7.770   | 9.227  |
| Province: Sud-Kivu                | 4.471  | 0.665   | 0.665   | 5.204  | 5.204   | 5.204   | 3.892  |
| Birth group: c 2-4 long           | 3.506  | 3.503   | 3.503   | -5.820 | -5.820  | -5.820  | -4.951 |
| Mother occupation: d Agriculture  | 3.463  | 4.521   | 4.521   | 15.488 | 15.488  | 15.488  | 15.615 |
| Province: Kwilu                   | 3.342  | 0.326   | 0.326   | -0.765 | -0.765  | -0.765  | -1.395 |
| Province: Nord-Kivu               | 2.859  | 0.243   | 0.243   | -0.323 | -0.323  | -0.323  | -0.002 |
| Partner education                 | 2.750  | 0.684   | 0.684   | -0.006 | -0.006  | -0.006  | 1.724  |

## 1.2.2 4.2.2 Root-Node Inequality

Table 2: Root-node inequality in under-five mortality by concentration-index criterion

| Root node   | CI     | CIg    | CIc    | L      |
|-------------|--------|--------|--------|--------|
| Full sample | 0.1112 | 0.0077 | 0.0308 | 0.0818 |

## 1.2.3 4.2.3 Selected Concentration-Index Tree

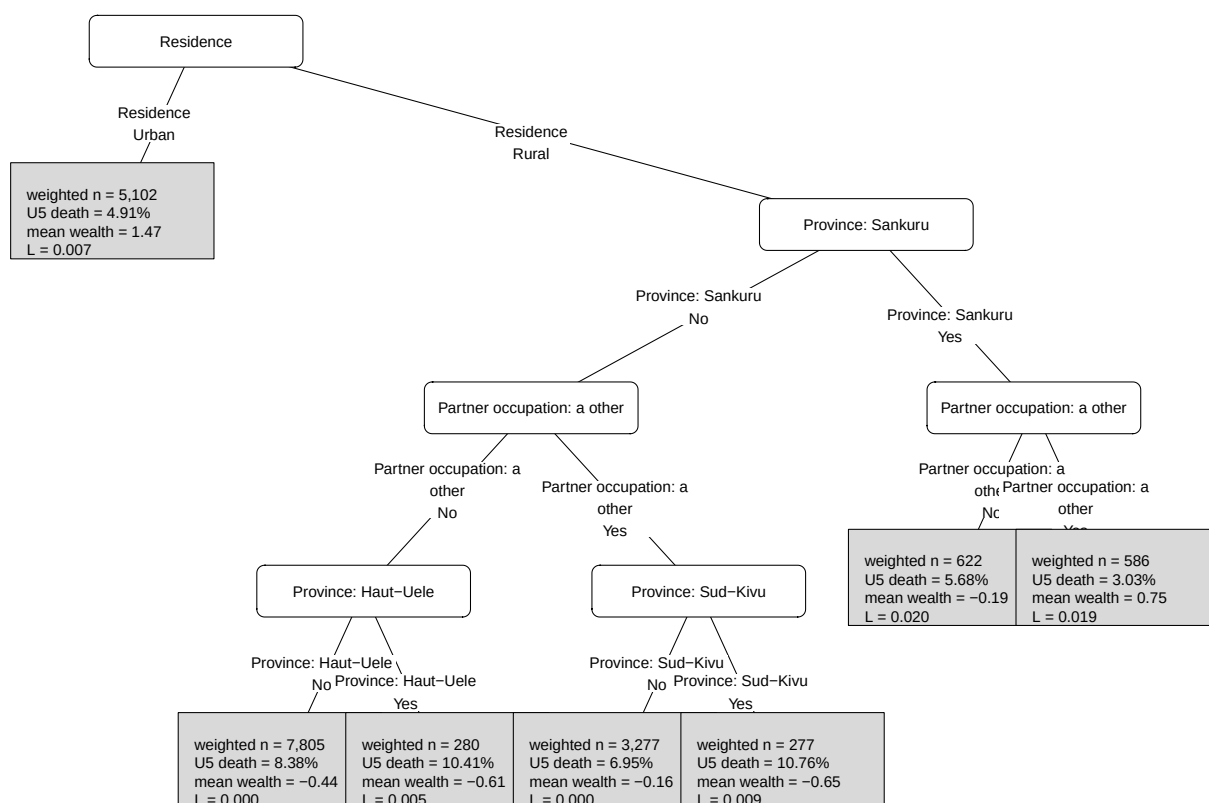


Figure 2: Selected concentration-index tree for under-five mortality in the DRC.

## 1.2.4 Surrogate Tree Comparison

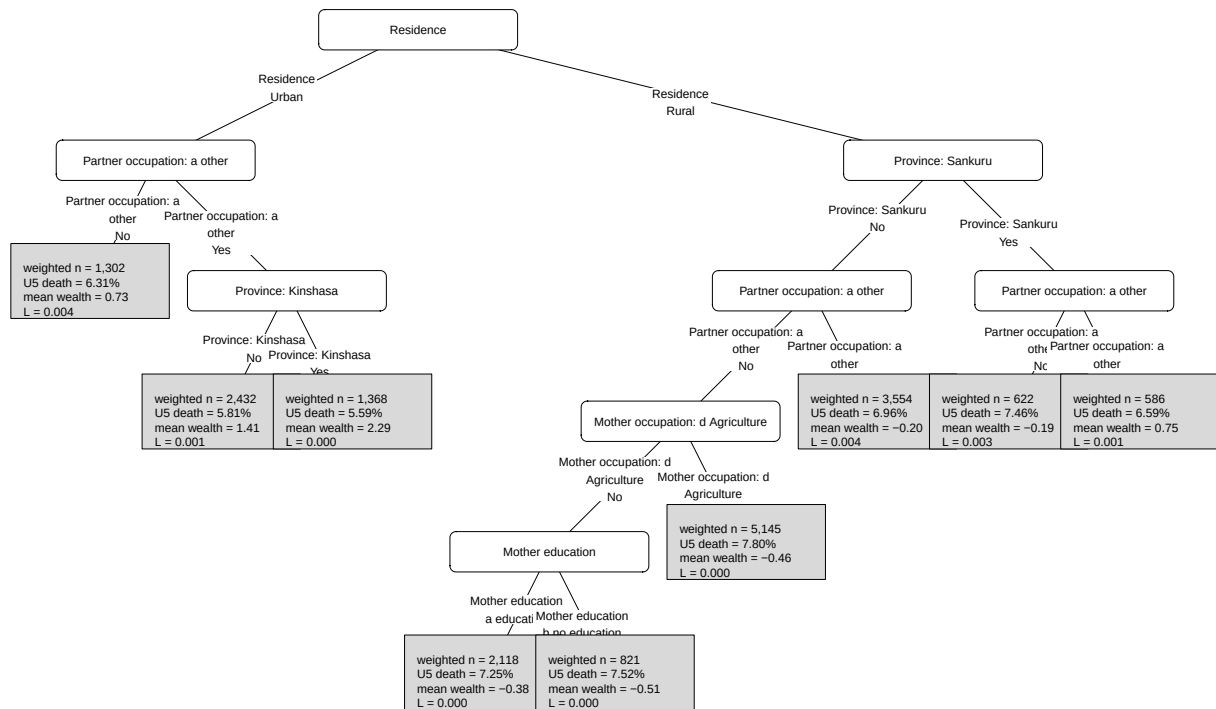


Figure 3: Surrogate tree summarising the selected concentration-index forest predictions.

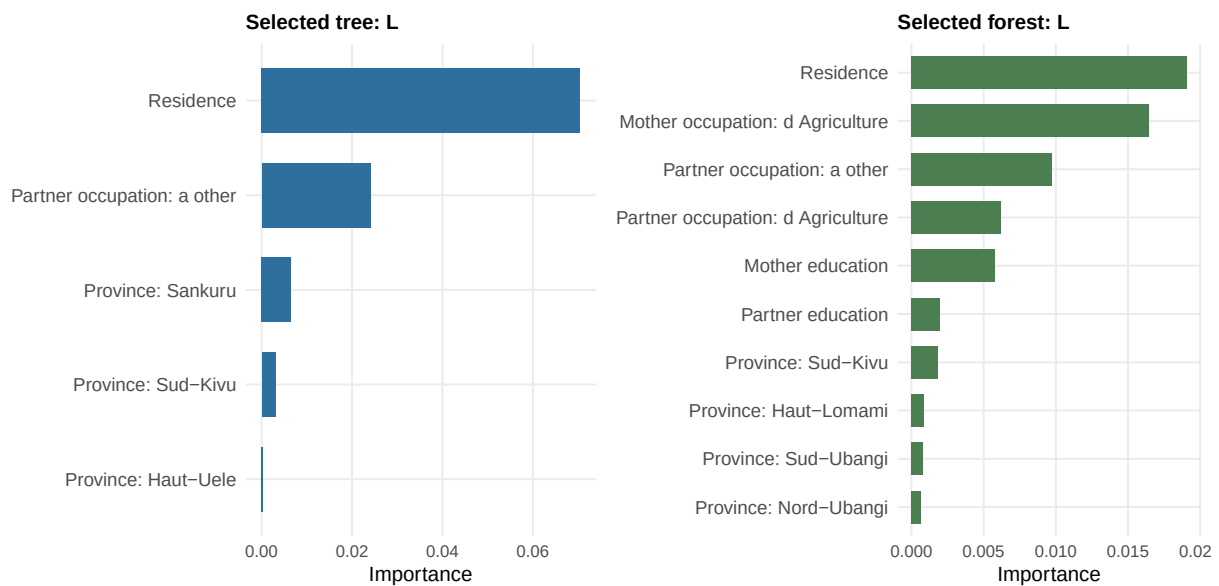


Figure 4: Variable importance for the selected concentration-index tree and forest.

## 1.3 4.3 Simulation Study

### 1.3.1 4.3.1 Recovery of the Simulated Subgroup Structure

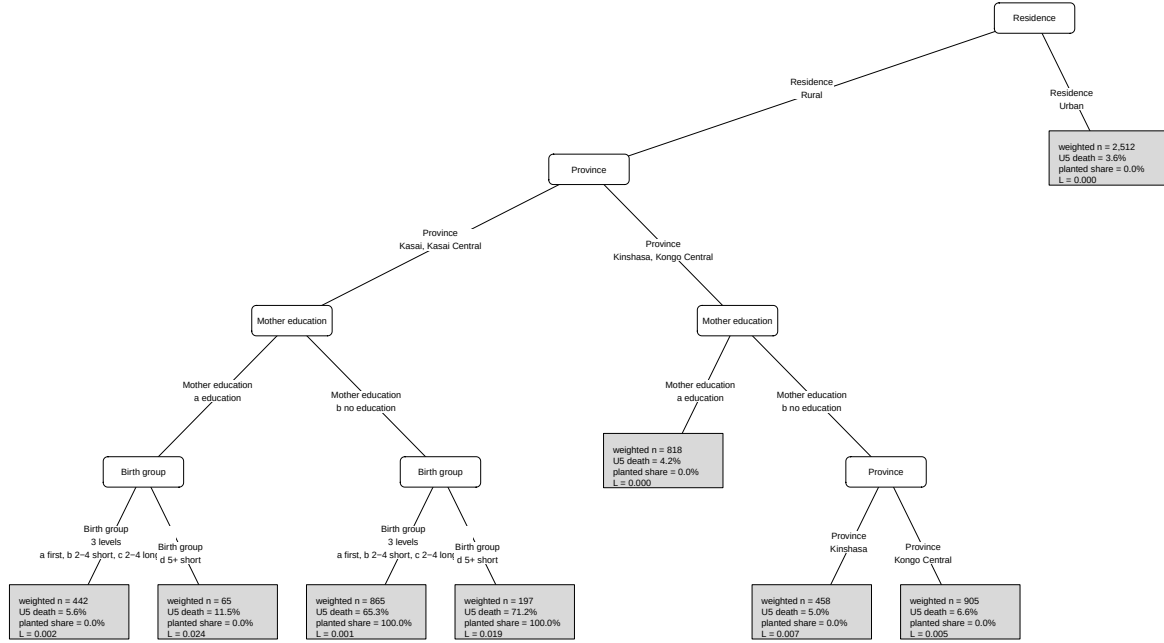


Figure 5: Tree recovered from simulated data with a known subgroup-generating mechanism.

## 2 Appendix

### 2.1 Results

#### 2.1.1 Appendix R1: Exploratory Data Outputs

Table 3: Weighted distribution of analysis variables by under-five mortality status

| Variable     | Under-five mortality |                    | Overall             |
|--------------|----------------------|--------------------|---------------------|
|              | Alive at age 5       | Died before age 5  | Overall             |
|              | (weighted N=16,704)  | (weighted N=1,244) | (weighted N=17,948) |
| Low-skill HH |                      |                    |                     |
| No           | 10,185.6 (61.0%)     | 786.1 (63.2%)      | 10,971.7 (61.1%)    |
| Yes          | 6,518.4 (39.0%)      | 458.2 (36.8%)      | 6,976.5 (38.9%)     |
| Child sex    |                      |                    |                     |
| Female       | 8,099.9 (48.5%)      | 515.2 (41.4%)      | 8,615.1 (48.0%)     |
| Male         | 8,604.1 (51.5%)      | 729 (58.6%)        | 9,333.1 (52.0%)     |
| Birth group  |                      |                    |                     |
| a first      | 2,691.8 (16.1%)      | 242.1 (19.5%)      | 2,933.9 (16.3%)     |
| b 2-4 short  | 2,112.9 (12.6%)      | 186.1 (15.0%)      | 2,299 (12.8%)       |
| c 2-4 long   | 5,861.4 (35.1%)      | 287 (23.1%)        | 6,148.4 (34.3%)     |
| d 5+ short   | 1,633.8 (9.8%)       | 248.3 (20.0%)      | 1,882.1 (10.5%)     |
| e 5+ long    | 4,404.1 (26.4%)      | 280.7 (22.6%)      | 4,684.8 (26.1%)     |

|                                            |                  |                 |                  |
|--------------------------------------------|------------------|-----------------|------------------|
| Mother age                                 |                  |                 |                  |
| a 20 or more                               | 16,100.6 (96.4%) | 1,172.9 (94.3%) | 17,273.4 (96.2%) |
| less than 20                               | 603.4 (3.6%)     | 71.4 (5.7%)     | 674.8 (3.8%)     |
| Residence                                  |                  |                 |                  |
| Urban                                      | 4,851.8 (29.0%)  | 250.3 (20.1%)   | 5,102.1 (28.4%)  |
| Rural                                      | 11,852.2 (71.0%) | 993.9 (79.9%)   | 12,846.1 (71.6%) |
| Mother education                           |                  |                 |                  |
| a education                                | 13,472.5 (80.7%) | 958 (77.0%)     | 14,430.5 (80.4%) |
| b no education                             | 3,231.5 (19.3%)  | 286.2 (23.0%)   | 3,517.7 (19.6%)  |
| Partner education                          |                  |                 |                  |
| a education                                | 15,311.1 (91.7%) | 1,106.1 (88.9%) | 16,417.2 (91.5%) |
| b no education                             | 1,392.9 (8.3%)   | 138.2 (11.1%)   | 1,531 (8.5%)     |
| Mother occupation                          |                  |                 |                  |
| a other                                    | 3,999.5 (23.9%)  | 254.8 (20.5%)   | 4,254.4 (23.7%)  |
| c Household, unskilled manual, not working | 5,307.9 (31.8%)  | 350 (28.1%)     | 5,657.9 (31.5%)  |
| d Agriculture                              | 7,396.6 (44.3%)  | 639.4 (51.4%)   | 8,036 (44.8%)    |
| Partner occupation                         |                  |                 |                  |
| a other                                    | 7,484.6 (44.8%)  | 455.9 (36.6%)   | 7,940.5 (44.2%)  |
| c Household, unskilled manual, not working | 2,704.3 (16.2%)  | 223 (17.9%)     | 2,927.3 (16.3%)  |
| d Agriculture                              | 6,515.1 (39.0%)  | 565.3 (45.4%)   | 7,080.4 (39.4%)  |

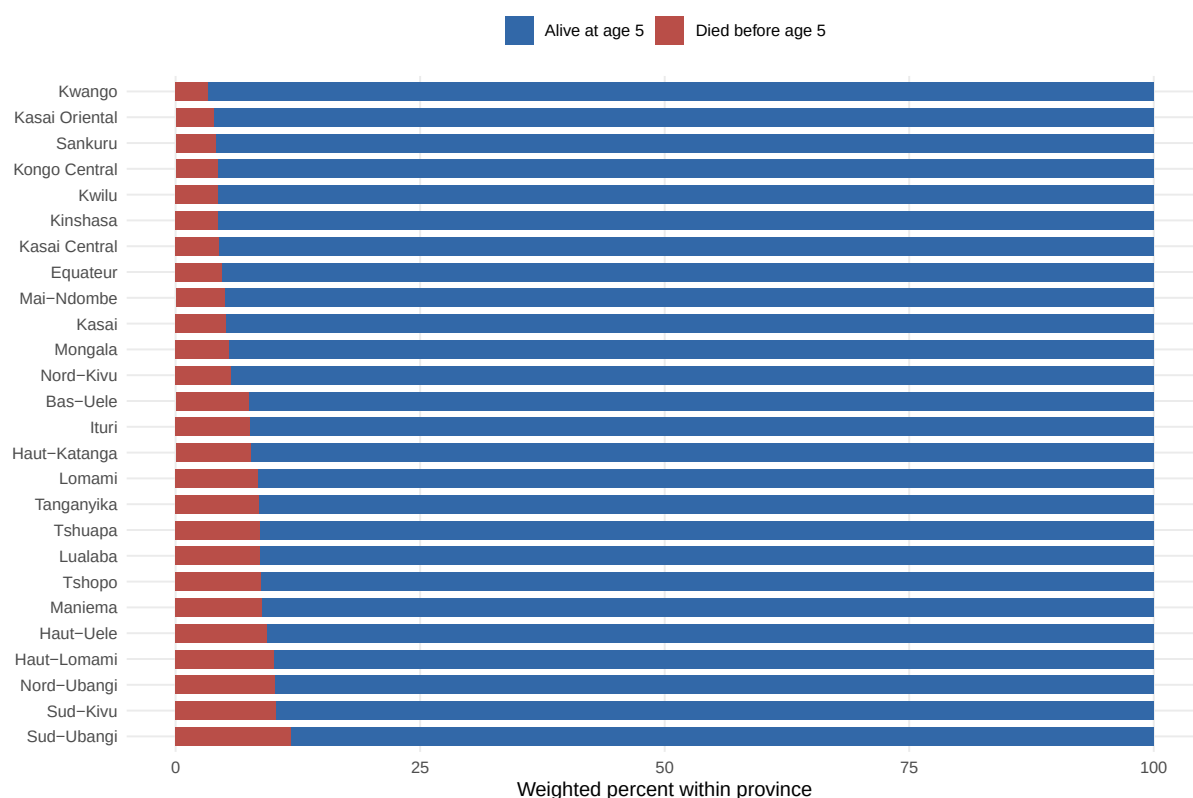


Figure 6: Weighted province distribution by under-five mortality status.

## 2.1.2 Appendix R2: Regression and Predictive-Forest Details

Table 4: Survey-weighted regression model summary

|  | Term                                        | Estimate | Std. error | Statistic | p-value | Conf. low | Conf. high |
|--|---------------------------------------------|----------|------------|-----------|---------|-----------|------------|
|  | Intercept                                   | -2.983   | 0.268      | -11.135   | < 0.001 | -3.509    | -2.458     |
|  | Low-skill HH                                | 0.009    | 0.241      | 0.037     | 0.97074 | -0.463    | 0.480      |
|  | Child sex                                   | 0.306    | 0.085      | 3.591     | < 0.001 | 0.139     | 0.472      |
|  | Birth group: b 2-4 short                    | -0.003   | 0.144      | -0.023    | 0.98128 | -0.285    | 0.278      |
|  | Birth group: c 2-4 long                     | -0.553   | 0.133      | -4.153    | < 0.001 | -0.814    | -0.292     |
|  | Birth group: d 5+ short                     | 0.434    | 0.146      | 2.980     | 0.00289 | 0.149     | 0.720      |
|  | Birth group: e 5+ long                      | -0.372   | 0.136      | -2.726    | 0.00642 | -0.639    | -0.104     |
|  | Mother age                                  | 0.319    | 0.178      | 1.787     | 0.07393 | -0.031    | 0.668      |
|  | Residence                                   | 0.384    | 0.137      | 2.799     | 0.00514 | 0.115     | 0.654      |
|  | Mother education                            | 0.039    | 0.113      | 0.345     | 0.73043 | -0.183    | 0.261      |
|  | Partner education                           | 0.199    | 0.151      | 1.319     | 0.18708 | -0.097    | 0.496      |
|  | Mother occupation: c                        | -0.087   | 0.223      | -0.390    | 0.69656 | -0.523    | 0.349      |
|  | Household, unskilled manual,<br>not working |          |            |           |         |           |            |
|  | Mother occupation: d                        | 0.051    | 0.131      | 0.391     | 0.69544 | -0.206    | 0.309      |
|  | Agriculture                                 |          |            |           |         |           |            |
|  | Partner occupation: c                       | 0.151    | 0.177      | 0.854     | 0.39300 | -0.196    | 0.498      |
|  | Household, unskilled manual,<br>not working |          |            |           |         |           |            |
|  | Partner occupation: d                       | 0.103    | 0.105      | 0.980     | 0.32707 | -0.103    | 0.309      |
|  | Agriculture                                 |          |            |           |         |           |            |
|  | Province: Kwango                            | -0.707   | 0.420      | -1.684    | 0.09218 | -1.530    | 0.116      |
|  | Province: Kwilu                             | -0.540   | 0.317      | -1.702    | 0.08886 | -1.162    | 0.082      |
|  | Province: Mai-Ndombe                        | -0.335   | 0.313      | -1.070    | 0.28473 | -0.949    | 0.279      |
|  | Province: Kongo Central                     | -0.420   | 0.454      | -0.926    | 0.35445 | -1.310    | 0.469      |
|  | Province: Equateur                          | -0.326   | 0.336      | -0.972    | 0.33098 | -0.984    | 0.331      |
|  | Province: Mongala                           | -0.273   | 0.312      | -0.877    | 0.38059 | -0.884    | 0.338      |
|  | Province: Nord-Ubangi                       | 0.302    | 0.302      | 0.999     | 0.31797 | -0.291    | 0.894      |
|  | Province: Sud-Ubangi                        | 0.489    | 0.336      | 1.457     | 0.14506 | -0.169    | 1.147      |
|  | Province: Tshuapa                           | 0.176    | 0.321      | 0.549     | 0.58318 | -0.453    | 0.806      |
|  | Province: Kasai                             | -0.303   | 0.335      | -0.906    | 0.36519 | -0.960    | 0.353      |
|  | Province: Kasai Central                     | -0.455   | 0.397      | -1.146    | 0.25179 | -1.232    | 0.323      |
|  | Province: Kasai Oriental                    | -0.615   | 0.606      | -1.016    | 0.30961 | -1.803    | 0.572      |
|  | Province: Lomami                            | 0.133    | 0.321      | 0.414     | 0.67890 | -0.496    | 0.761      |
|  | Province: Sankuru                           | -0.646   | 0.335      | -1.930    | 0.05366 | -1.302    | 0.010      |
|  | Province: Haut-Katanga                      | -0.002   | 0.315      | -0.007    | 0.99429 | -0.619    | 0.615      |
|  | Province: Haut-Lomami                       | 0.368    | 0.285      | 1.288     | 0.19765 | -0.192    | 0.927      |
|  | Province: Lualaba                           | 0.377    | 0.297      | 1.267     | 0.20508 | -0.206    | 0.959      |
|  | Province: Tanganyika                        | 0.154    | 0.292      | 0.528     | 0.59752 | -0.419    | 0.727      |
|  | Province: Maniema                           | 0.172    | 0.334      | 0.516     | 0.60586 | -0.482    | 0.826      |
|  | Province: Nord-Kivu                         | -0.368   | 0.347      | -1.058    | 0.29019 | -1.049    | 0.314      |
|  | Province: Bas-Uele                          | 0.137    | 0.291      | 0.470     | 0.63804 | -0.433    | 0.707      |
|  | Province: Haut-Uele                         | 0.197    | 0.323      | 0.611     | 0.54116 | -0.436    | 0.831      |
|  | Province: Ituri                             | 0.232    | 0.311      | 0.746     | 0.45549 | -0.378    | 0.842      |
|  | Province: Tshopo                            | 0.186    | 0.328      | 0.566     | 0.57124 | -0.457    | 0.829      |
|  | Province: Sud-Kivu                          | 0.335    | 0.292      | 1.146     | 0.25199 | -0.238    | 0.908      |

Table 5: Cross-validation results for the predictive ranger model

|  | Metric      | mtry | min_n | Mean   | Std. error | Folds | Rank |
|--|-------------|------|-------|--------|------------|-------|------|
|  | mn_log_loss | 1    | 150   | 0.2568 | 0.0038     | 5     | 1    |
|  | mn_log_loss | 1    | 80    | 0.2568 | 0.0038     | 5     | 2    |

|             |    |     |        |        |   |    |
|-------------|----|-----|--------|--------|---|----|
| mn_log_loss | 1  | 10  | 0.2568 | 0.0038 | 5 | 3  |
| mn_log_loss | 15 | 150 | 0.2606 | 0.0044 | 5 | 4  |
| mn_log_loss | 29 | 150 | 0.2630 | 0.0045 | 5 | 5  |
| mn_log_loss | 43 | 150 | 0.2643 | 0.0046 | 5 | 6  |
| mn_log_loss | 15 | 80  | 0.2656 | 0.0047 | 5 | 7  |
| mn_log_loss | 29 | 80  | 0.2693 | 0.0051 | 5 | 8  |
| mn_log_loss | 43 | 80  | 0.2781 | 0.0086 | 5 | 9  |
| mn_log_loss | 15 | 10  | 0.2964 | 0.0058 | 5 | 10 |
| mn_log_loss | 29 | 10  | 0.3650 | 0.0074 | 5 | 11 |
| mn_log_loss | 43 | 10  | 0.4987 | 0.0208 | 5 | 12 |
| roc_auc     | 1  | 80  | 0.6270 | 0.0105 | 5 | 1  |
| roc_auc     | 1  | 10  | 0.6269 | 0.0103 | 5 | 2  |
| roc_auc     | 1  | 150 | 0.6262 | 0.0110 | 5 | 3  |
| roc_auc     | 15 | 150 | 0.6103 | 0.0068 | 5 | 4  |
| roc_auc     | 29 | 150 | 0.6066 | 0.0077 | 5 | 5  |
| roc_auc     | 43 | 150 | 0.6055 | 0.0081 | 5 | 6  |
| roc_auc     | 15 | 80  | 0.6044 | 0.0074 | 5 | 7  |
| roc_auc     | 29 | 80  | 0.6004 | 0.0086 | 5 | 8  |
| roc_auc     | 43 | 80  | 0.5972 | 0.0093 | 5 | 9  |
| roc_auc     | 15 | 10  | 0.5824 | 0.0068 | 5 | 10 |
| roc_auc     | 29 | 10  | 0.5738 | 0.0072 | 5 | 11 |
| roc_auc     | 43 | 10  | 0.5702 | 0.0076 | 5 | 12 |



Figure 7: SHAP summary plot for the predictive ranger model.

### 2.1.3 Appendix R3: Concentration-Index Tree Details

Table 6: Best tree hyperparameters selected by validation gain within each impurity criterion

| type | minsplit | minbucket | minprob | maxdepth | min_gain | min_relative_gain |
|------|----------|-----------|---------|----------|----------|-------------------|
| CI   | 500      | 250       | 0.01    | 10       | 0        | 0.2               |
| CIg  | 500      | 250       | 0.01    | 10       | 0        | 0.2               |
| CIc  | 500      | 250       | 0.01    | 10       | 0        | 0.2               |
| L    | 500      | 250       | 0.01    | 10       | 0        | 0.2               |

Table 7: Tree summary for the selected settings

| criterion | metric                   | estimate | std_error |
|-----------|--------------------------|----------|-----------|
| CI        | Root impurity            | 0.1115   | NA        |
| CI        | Training gain            | 0.0505   | 0.0030    |
| CI        | Training gain (% root)   | 45.5193  | 2.7884    |
| CI        | Validation gain          | 0.0067   | 0.0122    |
| CI        | Validation gain (% root) | -9.2483  | 14.5979   |
| CIc       | Root impurity            | 0.0306   | NA        |
| CIc       | Training gain            | 0.0190   | 0.0008    |
| CIc       | Training gain (% root)   | 61.6639  | 2.4017    |
| CIc       | Validation gain          | -0.0003  | 0.0062    |
| CIc       | Validation gain (% root) | -37.2283 | 30.9978   |
| CIg       | Root impurity            | 0.0076   | NA        |
| CIg       | Training gain            | 0.0048   | 0.0002    |
| CIg       | Training gain (% root)   | 61.6639  | 2.4017    |
| CIg       | Validation gain          | -0.0001  | 0.0015    |
| CIg       | Validation gain (% root) | -37.2283 | 30.9978   |
| L         | Root impurity            | 0.0878   | NA        |
| L         | Training gain            | 0.0775   | 0.0013    |
| L         | Training gain (% root)   | 94.7077  | 0.3721    |
| L         | Validation gain          | 0.0641   | 0.0184    |
| L         | Validation gain (% root) | 64.4396  | 9.8198    |

Table 8: Selection metrics for fitted concentration-index trees

| Criterion | Root objective | Train gain | Train gain/root | Validation gain | Validation gain/root | Terminal nodes |
|-----------|----------------|------------|-----------------|-----------------|----------------------|----------------|
| CI        | 0.1115         | 0.0505     | 0.4552          | 0.0067          | -0.0925              | 3.9            |
| CIc       | 0.0306         | 0.0190     | 0.6166          | -0.0003         | -0.3723              | 6.1            |
| CIg       | 0.0076         | 0.0048     | 0.6166          | -0.0001         | -0.3723              | 6.1            |
| L         | 0.0878         | 0.0775     | 0.9471          | 0.0641          | 0.6444               | 6.0            |

Table 9: Collected tree complexity metrics from cross-validation

| type | maxdepth | min_relative_gain | mean_terminal_nodes | mean_train_relative_gain | mean_validation |
|------|----------|-------------------|---------------------|--------------------------|-----------------|
| CI   | 10       | 0.05              | 14.3                |                          | 0.6752          |
| CI   | 10       | 0.10              | 12.0                |                          | 0.6557          |
| CI   | 10       | 0.20              | 3.9                 |                          | 0.4552          |
| CIc  | 10       | 0.05              | 14.7                |                          | 0.7590          |
| CIc  | 10       | 0.10              | 12.8                |                          | 0.7439          |
| CIc  | 10       | 0.20              | 6.1                 |                          | 0.6166          |

|     |    |      |      |        |
|-----|----|------|------|--------|
| CIg | 10 | 0.05 | 14.7 | 0.7590 |
| CIg | 10 | 0.10 | 12.8 | 0.7439 |
| CIg | 10 | 0.20 | 6.1  | 0.6166 |
| L   | 10 | 0.05 | 10.6 | 0.9582 |
| L   | 10 | 0.10 | 9.2  | 0.9558 |
| L   | 10 | 0.20 | 6.0  | 0.9471 |

Table 10: Average tree validation percentage and

| type | min_relative_gain | n_settings | mean_root_objective | mean_validation_gain | validation_percent_ |
|------|-------------------|------------|---------------------|----------------------|---------------------|
| CI   | 0.05              | 1          | 0.1115              | -0.0833              |                     |
| CI   | 0.10              | 1          | 0.1115              | -0.0691              |                     |
| CI   | 0.20              | 1          | 0.1115              | 0.0067               |                     |
| CIc  | 0.05              | 1          | 0.0306              | -0.0169              |                     |
| CIc  | 0.10              | 1          | 0.0306              | -0.0153              |                     |
| CIc  | 0.20              | 1          | 0.0306              | -0.0003              |                     |
| CIg  | 0.05              | 1          | 0.0076              | -0.0042              |                     |
| CIg  | 0.10              | 1          | 0.0076              | -0.0038              |                     |
| CIg  | 0.20              | 1          | 0.0076              | -0.0001              |                     |
| L    | 0.05              | 1          | 0.0878              | 0.0601               |                     |
| L    | 0.10              | 1          | 0.0878              | 0.0602               |                     |
| L    | 0.20              | 1          | 0.0878              | 0.0641               |                     |

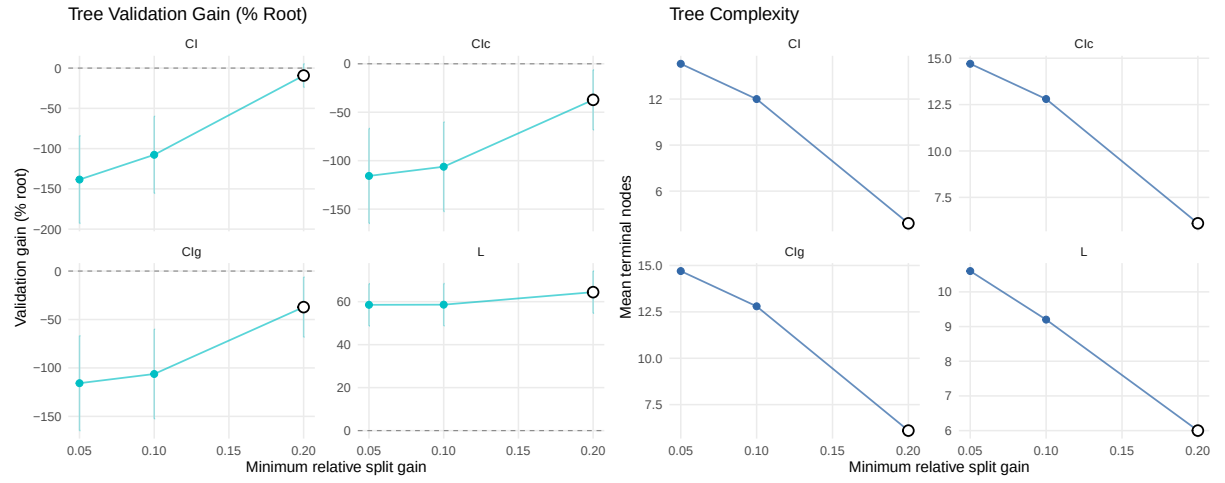


Figure 8: Tree validation gain and complexity by minimum relative split gain.

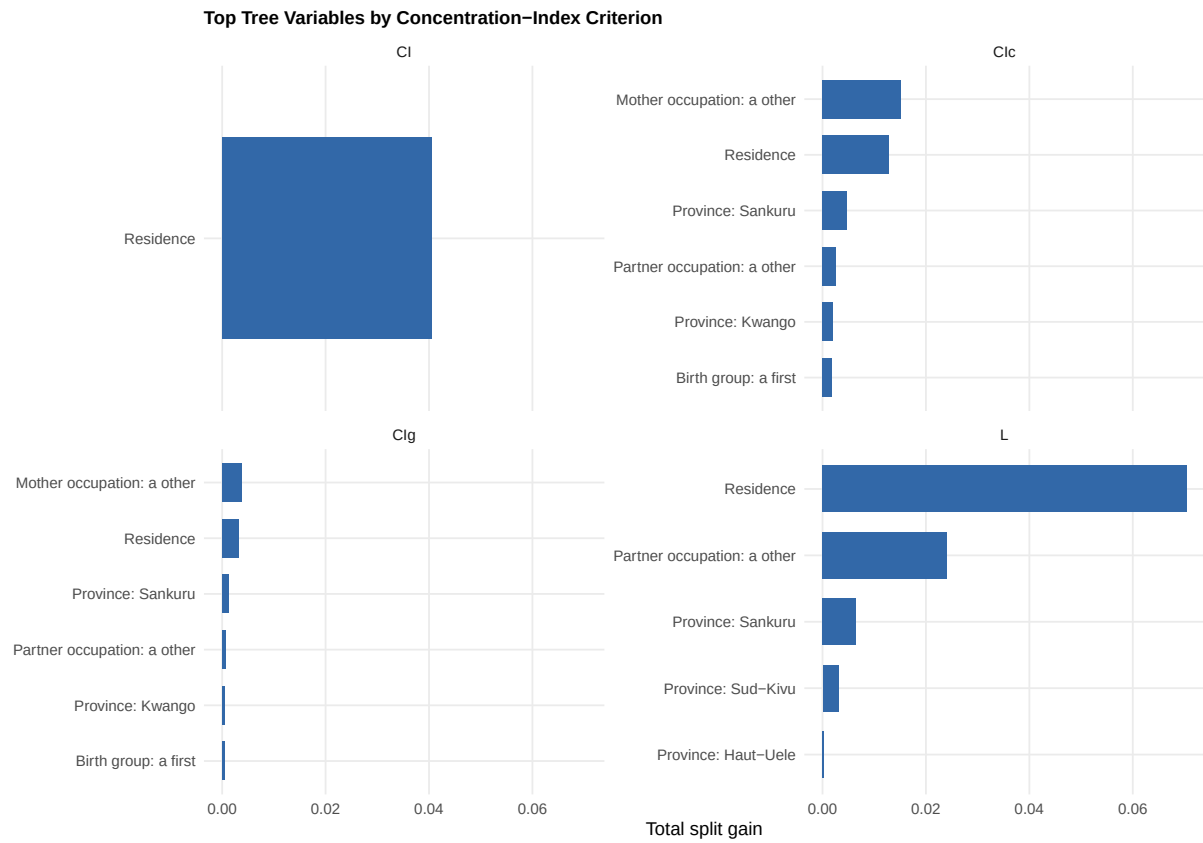


Figure 9: Top tree variables by concentration-index criterion.

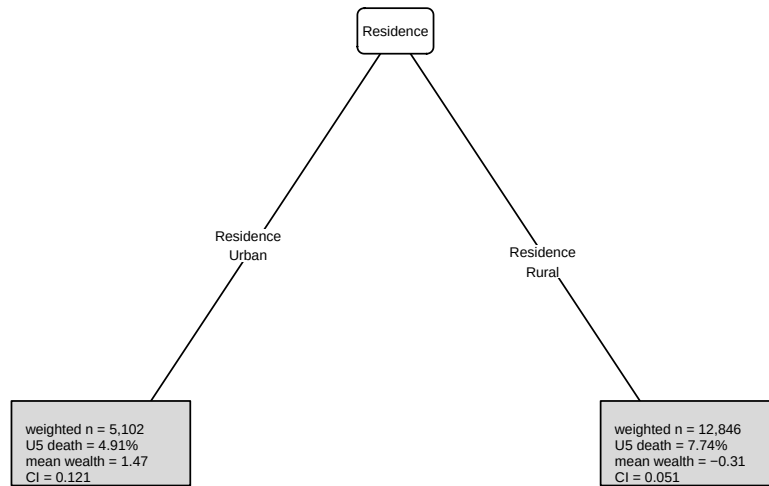


Figure 10: Concentration-index tree fitted with the CI criterion.

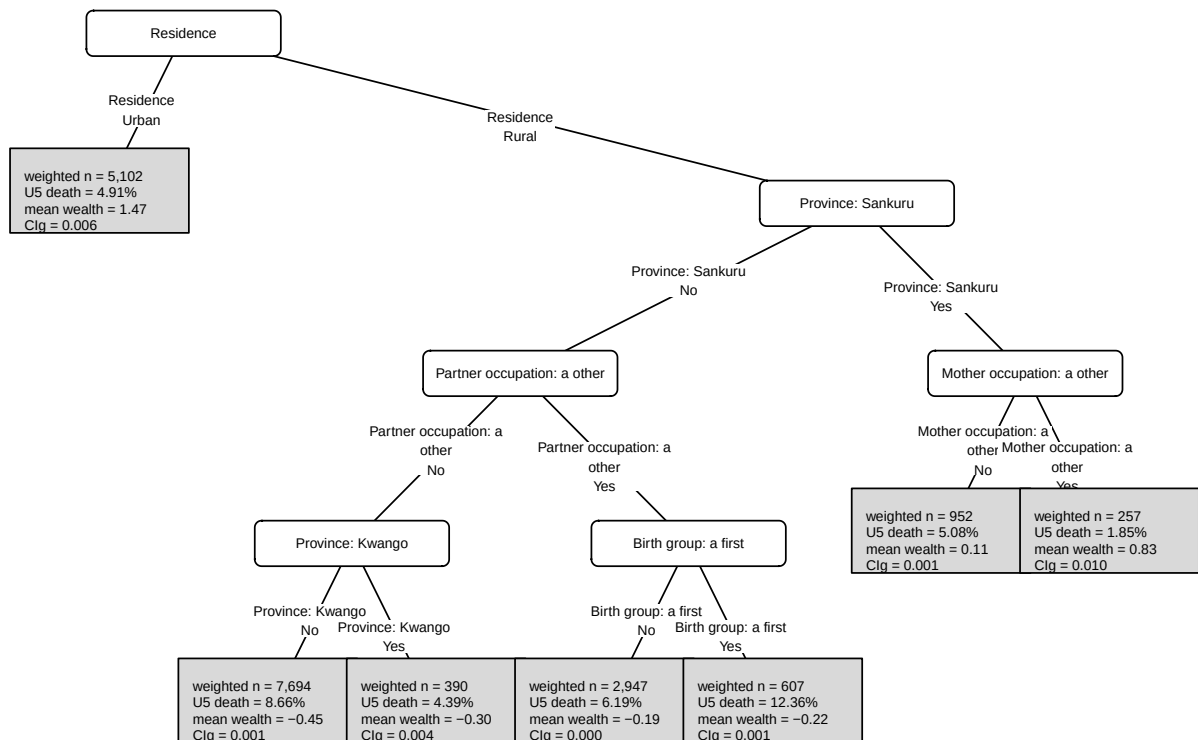


Figure 11: Concentration-index tree fitted with the CIg criterion.

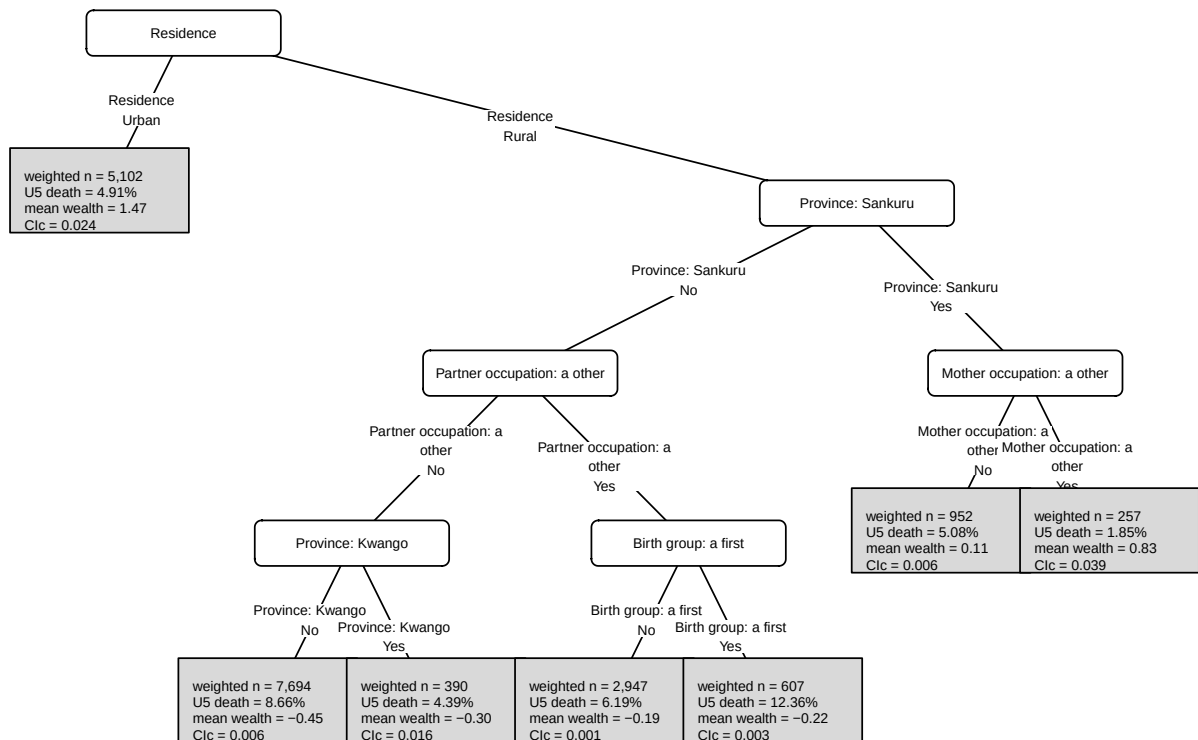


Figure 12: Concentration-index tree fitted with the CIc criterion.

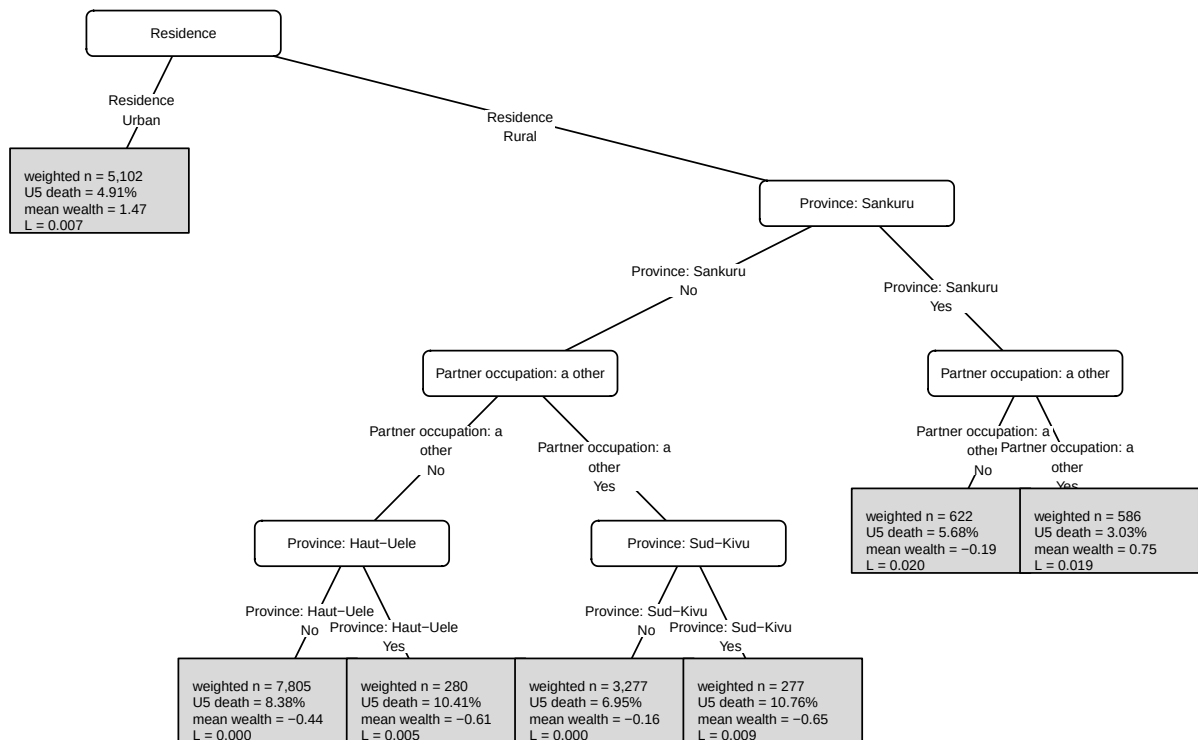


Figure 13: Concentration-index tree fitted with the L criterion.

### 2.1.4 Appendix R4: Concentration-Index Forest Details

Table 11: Best forest hyperparameters selected by validation gain within each impurity criterion

| type | ntree | mtry | minsplit | minbucket | minprob | maxdepth | min_gain | min_relative_gain |
|------|-------|------|----------|-----------|---------|----------|----------|-------------------|
| CI   | 100   | 6    | 500      | 250       | 0.01    | 10       | 0        | 0.30              |
| CIc  | 100   | 6    | 500      | 250       | 0.01    | 10       | 0        | 0.30              |
| CIg  | 100   | 6    | 500      | 250       | 0.01    | 10       | 0        | 0.30              |
| L    | 100   | 6    | 500      | 250       | 0.01    | 10       | 0        | 0.05              |

Table 12: Forest summary for the selected settings

| criterion | metric                   | estimate | std_error |
|-----------|--------------------------|----------|-----------|
| CI        | Root impurity            | 0.1115   | NA        |
| CI        | Training gain            | 0.0049   | 0.0008    |
| CI        | Training gain (% root)   | 4.3665   | 0.6793    |
| CI        | Validation gain          | 0.0009   | 0.0017    |
| CI        | Validation gain (% root) | -2.0077  | 2.6558    |
| CIc       | Root impurity            | 0.0306   | NA        |
| CIc       | Training gain            | 0.0023   | 0.0003    |
| CIc       | Training gain (% root)   | 7.6127   | 0.8394    |
| CIc       | Validation gain          | 0.0007   | 0.0008    |
| CIc       | Validation gain (% root) | -2.0691  | 4.5030    |
| CIg       | Root impurity            | 0.0076   | NA        |
| CIg       | Training gain            | 0.0005   | 0.0000    |
| CIg       | Training gain (% root)   | 6.5594   | 0.5938    |
| CIg       | Validation gain          | 0.0002   | 0.0002    |
| CIg       | Validation gain (% root) | -1.5263  | 4.1046    |
| L         | Root impurity            | 0.0878   | NA        |
| L         | Training gain            | 0.0586   | 0.0015    |
| L         | Training gain (% root)   | 71.5086  | 0.6721    |
| L         | Validation gain          | 0.0507   | 0.0136    |
| L         | Validation gain (% root) | 47.7365  | 7.7605    |

Table 13: Selection metrics for fitted concentration-index forests

| Criterion | Trees | mtry | Root objective | Train gain | Train gain/root | Validation gain | Validation gain/root |
|-----------|-------|------|----------------|------------|-----------------|-----------------|----------------------|
| CI        | 100   | 6    | 0.1115         | 0.0049     | 0.0437          | 0.0009          | -0.02                |
| CIc       | 100   | 6    | 0.0306         | 0.0023     | 0.0761          | 0.0007          | -0.02                |
| CIg       | 100   | 6    | 0.0076         | 0.0005     | 0.0656          | 0.0002          | -0.01                |
| L         | 100   | 6    | 0.0878         | 0.0586     | 0.7151          | 0.0507          | 0.47                 |

Table 14: Collected forest complexity metrics from cross-validation

| type | ntree | maxdepth | min_relative_gain | mean_terminal_nodes | mean_train_relative_gain | mean_validation_relative_gain |
|------|-------|----------|-------------------|---------------------|--------------------------|-------------------------------|
| CI   | 100   | 10       | 0.05              | 3.107               | 0.1935                   | 0.1935                        |
| CI   | 100   | 10       | 0.10              | 2.080               | 0.1451                   | 0.1451                        |
| CI   | 100   | 10       | 0.20              | 1.310               | 0.0743                   | 0.0743                        |
| CI   | 100   | 10       | 0.30              | 1.133               | 0.0437                   | 0.0437                        |
| CIc  | 100   | 10       | 0.05              | 3.299               | 0.2552                   | 0.2552                        |

|     |     |    |      |       |        |
|-----|-----|----|------|-------|--------|
| CIc | 100 | 10 | 0.10 | 2.146 | 0.1834 |
| CIc | 100 | 10 | 0.20 | 1.549 | 0.1291 |
| CIc | 100 | 10 | 0.30 | 1.244 | 0.0761 |
| CIg | 100 | 10 | 0.05 | 3.271 | 0.2497 |
| CIg | 100 | 10 | 0.10 | 2.184 | 0.1928 |
| CIg | 100 | 10 | 0.20 | 1.510 | 0.1265 |
| CIg | 100 | 10 | 0.30 | 1.213 | 0.0656 |
| L   | 100 | 10 | 0.05 | 3.659 | 0.7151 |
| L   | 100 | 10 | 0.10 | 2.766 | 0.6021 |
| L   | 100 | 10 | 0.20 | 1.992 | 0.4923 |
| L   | 100 | 10 | 0.30 | 1.680 | 0.4063 |

Table 15: Average forest validation percentage and

| type | min_relative_gain | n_settings | mean_root_objective | mean_validation_gain | validation_percent |
|------|-------------------|------------|---------------------|----------------------|--------------------|
| CI   | 0.05              | 1          | 0.1115              | -0.0160              |                    |
| CI   | 0.10              | 1          | 0.1115              | -0.0051              |                    |
| CI   | 0.20              | 1          | 0.1115              | 0.0003               |                    |
| CI   | 0.30              | 1          | 0.1115              | 0.0009               |                    |
| CIc  | 0.05              | 1          | 0.0306              | -0.0020              |                    |
| CIc  | 0.10              | 1          | 0.0306              | -0.0007              |                    |
| CIc  | 0.20              | 1          | 0.0306              | 0.0001               |                    |
| CIc  | 0.30              | 1          | 0.0306              | 0.0007               |                    |
| CIg  | 0.05              | 1          | 0.0076              | -0.0006              |                    |
| CIg  | 0.10              | 1          | 0.0076              | -0.0001              |                    |
| CIg  | 0.20              | 1          | 0.0076              | 0.0000               |                    |
| CIg  | 0.30              | 1          | 0.0076              | 0.0002               |                    |
| L    | 0.05              | 1          | 0.0878              | 0.0507               |                    |
| L    | 0.10              | 1          | 0.0878              | 0.0454               |                    |
| L    | 0.20              | 1          | 0.0878              | 0.0381               |                    |
| L    | 0.30              | 1          | 0.0878              | 0.0306               |                    |

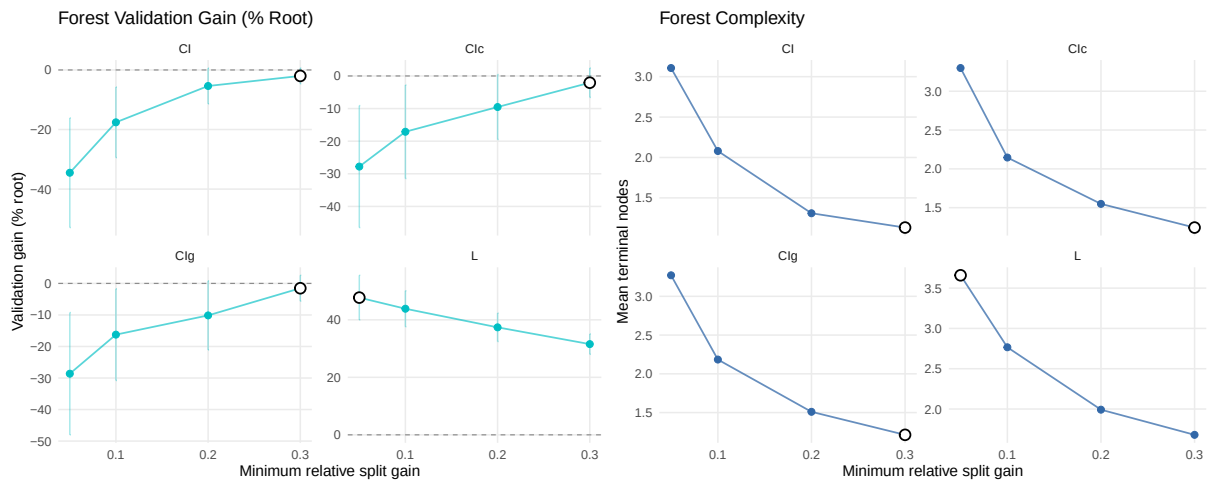


Figure 14: Forest validation gain and complexity by minimum relative split gain.

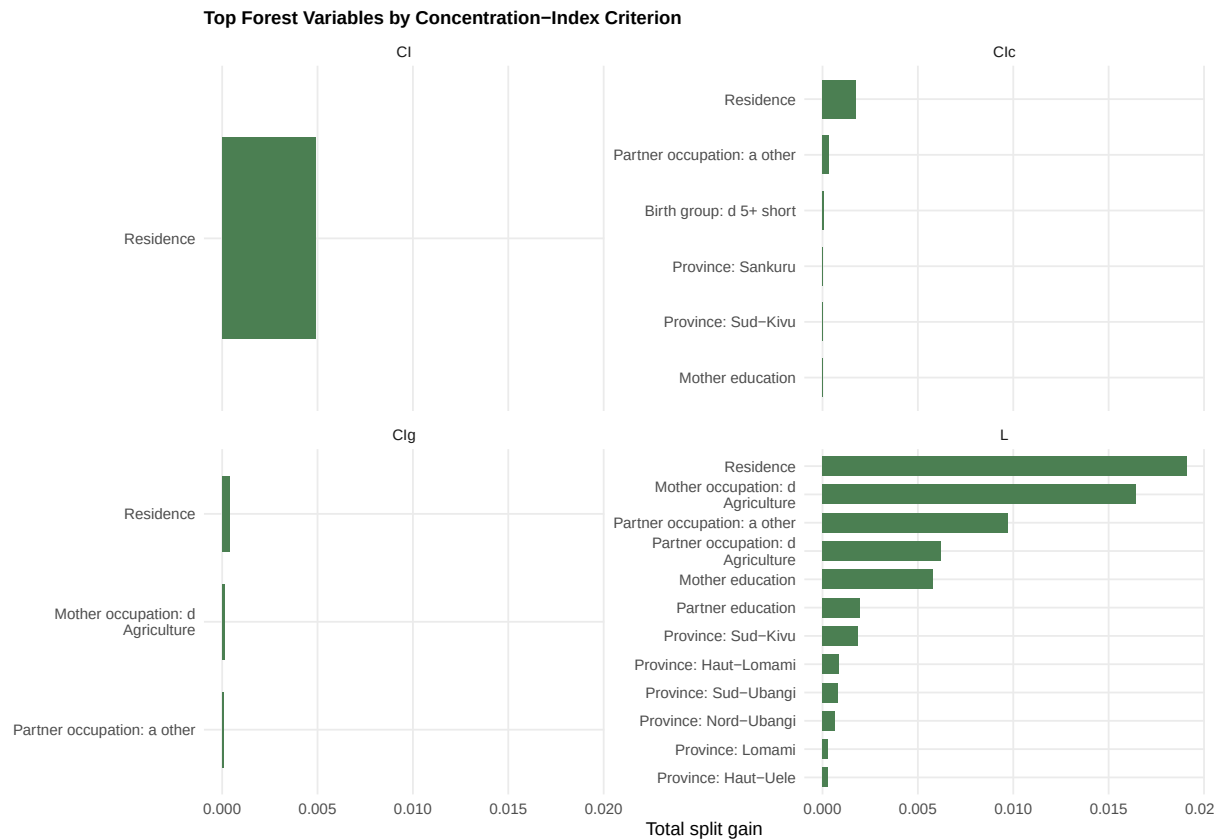


Figure 15: Top forest variables by concentration-index criterion.

### Non-Selected Forest Surrogate Trees

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

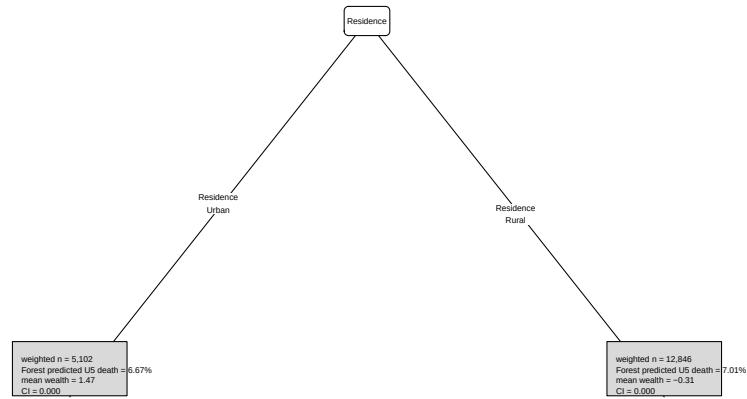


Figure 16: Surrogate tree for the CI concentration-index forest.

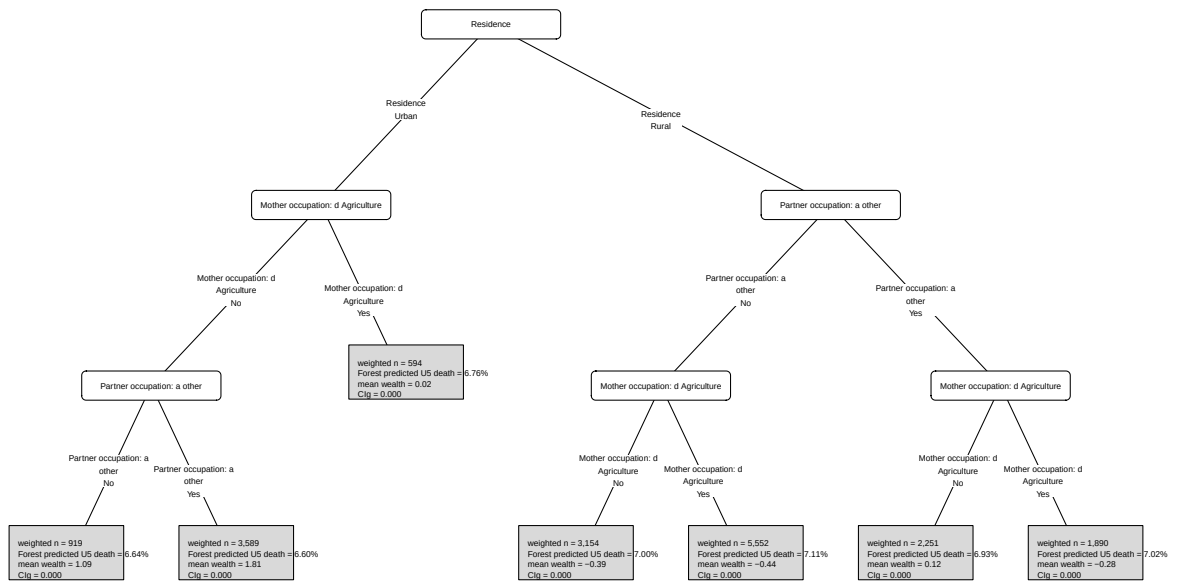


Figure 17: Surrogate tree for the CIg concentration-index forest.

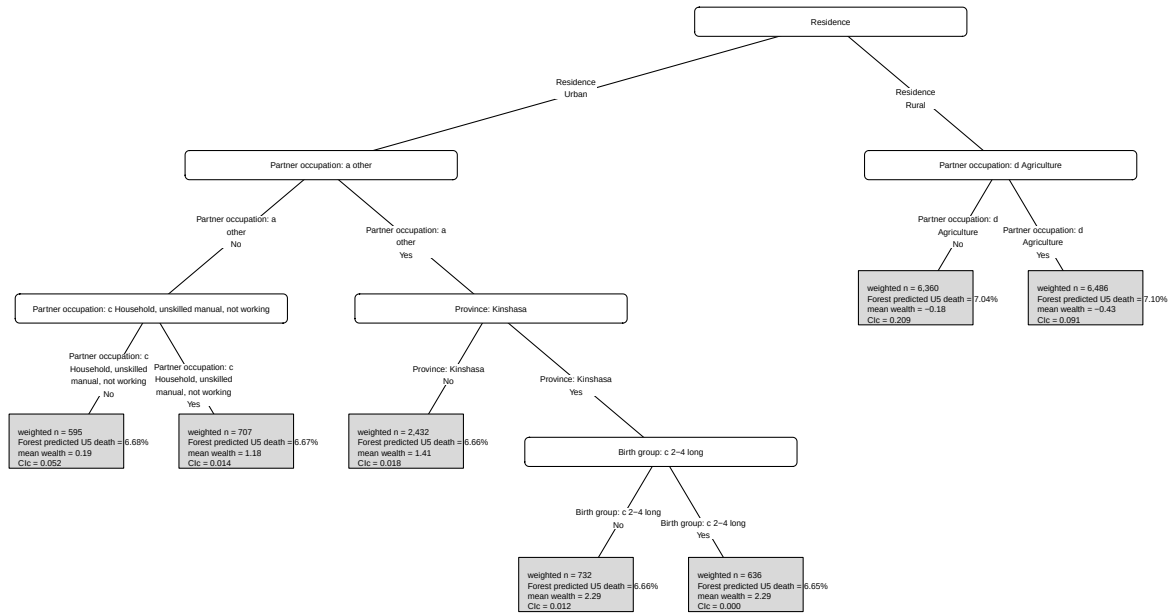


Figure 18: Surrogate tree for the CIc concentration-index forest.

Table 16: Forest variable importance by concentration-index criterion

| criterion | rank | variable                          | importance |
|-----------|------|-----------------------------------|------------|
| CI        | 1    | Residence                         | 0.0049     |
| CIc       | 1    | Residence                         | 0.0018     |
| CIc       | 2    | Partner occupation: a other       | 0.0003     |
| CIc       | 3    | Birth group: d 5+ short           | 0.0001     |
| CIc       | 4    | Province: Sankuru                 | 0.0000     |
| CIc       | 5    | Province: Sud-Kivu                | 0.0000     |
| CIc       | 6    | Mother education                  | 0.0000     |
| CIg       | 1    | Residence                         | 0.0004     |
| CIg       | 2    | Mother occupation: d Agriculture  | 0.0001     |
| CIg       | 3    | Partner occupation: a other       | 0.0001     |
| L         | 1    | Residence                         | 0.0191     |
| L         | 2    | Mother occupation: d Agriculture  | 0.0164     |
| L         | 3    | Partner occupation: a other       | 0.0097     |
| L         | 4    | Partner occupation: d Agriculture | 0.0062     |
| L         | 5    | Mother education                  | 0.0058     |
| L         | 6    | Partner education                 | 0.0020     |
| L         | 7    | Province: Sud-Kivu                | 0.0018     |
| L         | 8    | Province: Haut-Lomami             | 0.0009     |
| L         | 9    | Province: Sud-Ubangi              | 0.0008     |
| L         | 10   | Province: Nord-Ubangi             | 0.0007     |
| L         | 11   | Province: Lomami                  | 0.0003     |
| L         | 12   | Province: Haut-Uele               | 0.0003     |

Table 17: Surrogate-tree variable importance for the selected concentration-index forest

| rank | variable                         | importance |
|------|----------------------------------|------------|
| 1    | Residence                        | 0.0323     |
| 2    | Partner occupation: a other      | 0.0087     |
| 3    | Province: Sankuru                | 0.0009     |
| 4    | Province: Kinshasa               | 0.0002     |
| 5    | Mother education                 | 0.0002     |
| 6    | Mother occupation: d Agriculture | 0.0001     |

### 2.1.5 Appendix R5: Implementation Status

Implemented in the report: selected concentration-index tree and selected forest/surrogate-tree variable-importance plots in the results section; tree and concentration-index forest best-control tables, selected-setting validation summaries, complexity metrics, validation-path plots, and variable-importance outputs by criterion in the appendix; ranger cross-validation and SHAP summary outputs in the appendix.

To implement for the final analysis run: rerun the generator with `RUN_CV=true` and `RUN_FOREST_CV=true` when the full cross-validation objects should be recomputed from scratch rather than read from previously generated results.